

Do cluster-validity criteria recover the number of market regimes?

360 synthetic series | 4 nulls | a century of US returns

1. Both controls pass

neither the harness nor the criterion code is at fault

ARI = 1.00

clustering the latent volatility σ_t
recovers the true partition exactly

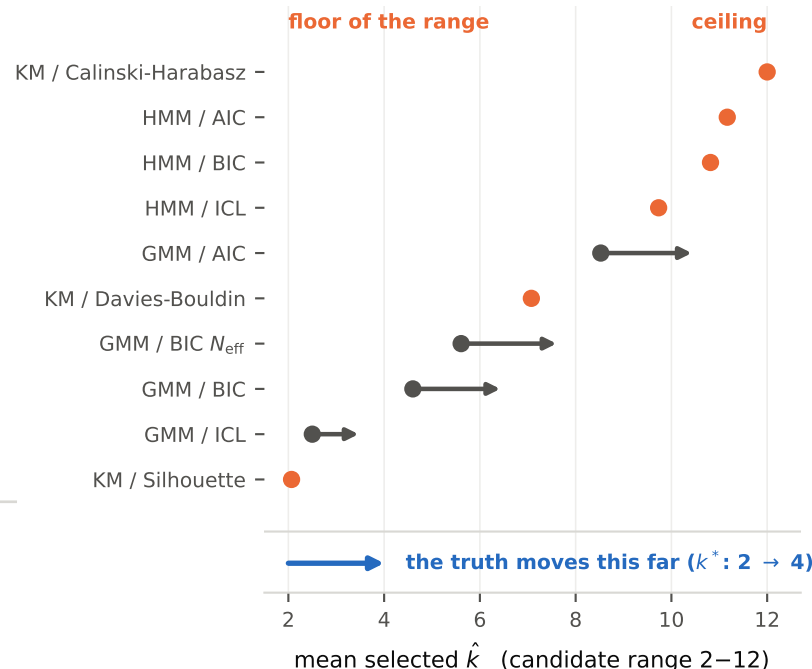
100% correct

the same Silhouette, Davies–Bouldin and
Calinski–Harabasz code recovers k^* on
well-separated Gaussian clusters

on the volatility feature the same harness
falls to ARI 0.25 at the standard proxy

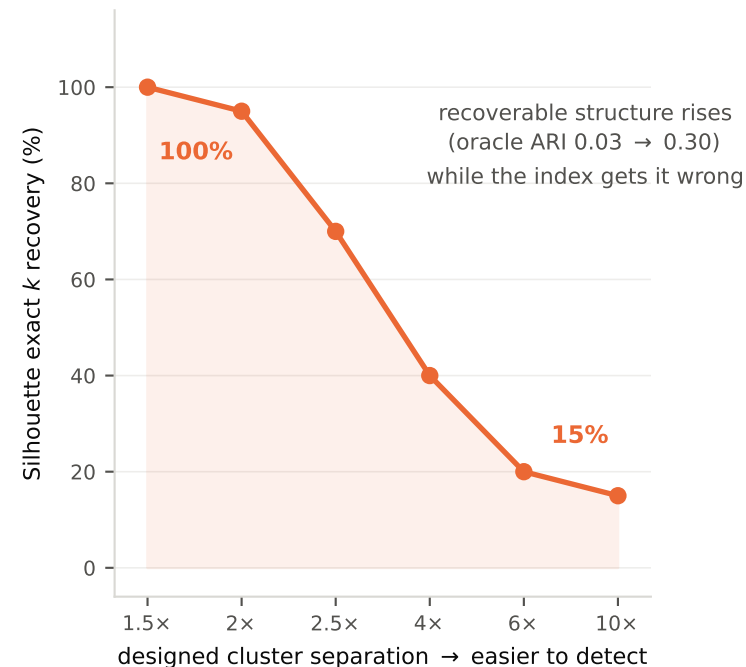
2. Yet the answers pin to a boundary

and six of the ten never move when the truth does



3. And accuracy runs backwards

right only while the clusters stay invisible



A criterion pinned to its own candidate boundary will corroborate whatever k the analyst brings to it. Report the controls that make the pinning visible.